Cosc 30: Discrete Mathematics

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Problem 1. We are playing a solitary game by repeatedly flipping a fair coin and record the results. For each of the following property, compute the expected number of coin flips until that property holds for the flip sequence using random variables.

1. Two heads appear
2. Both a head and a tail have appeared (on different flips)
3. Two consecutive heads appear

Solution.

1. Let E_{2H} be the number of expected flips for two heads to appear, E_H be the number of expected flips for a heads appears starting with 0 heads, and E_{1H} be the number of expected flips for a heads appears given we have already seen a heads. Then,

$$E_{2H} = \frac{1}{2}(1 + E_H) + \frac{1}{2}(1 + E_{2H})$$

Since there is 0.5 probability that we get a heads on the first flip, E_H is,

$$\begin{aligned} E_H &= \frac{1}{2} + \frac{1}{2}(1 + E_H) \\ \frac{1}{2}E_H &= 1 \\ E_H &= 2 \end{aligned}$$

And since the event of getting another heads given we have already seen a heads is the same as getting 1 head given we have 0 seen heads,

$$\begin{aligned}
E_{1H} &= \frac{1}{2} + \frac{1}{2}(1 + E_{1H}) \\
E_{1H} &= 1 + \frac{1}{2}E_{1H} \\
\frac{1}{2}E_{1H} &= 1 \\
E_{1H} &= 2
\end{aligned}$$

Then,

$$\begin{aligned}
E_{2H} &= \frac{1}{2}(1 + E_{H}) + \frac{1}{2}(1 + E_{2H}) \\
E_{2H} &= \frac{1}{2}(1 + 2) + \frac{1}{2}(1 + E_{2H}) \\
E_{2H} &= \frac{4}{2} + \frac{1}{2}(E_{2H}) \\
\frac{1}{2}E_{2H} &= 2 \\
E_{2H} &= 4
\end{aligned}$$

So the expected value E_{2H} for getting two heads is 4 flips.

2. Let E_{HT} be the number of expected flips for both heads and tails to appear, E_H be the number of expected flips for a heads to appear, and E_T be the number of expected flips for a tails to appear. Then,

$$E_{HT} = \frac{1}{2}(1 + E_H) + \frac{1}{2}(1 + E_T)$$

Since there is 0.5 probability that we get a heads on the first flip, E_H is,

$$\begin{aligned}
E_H &= \frac{1}{2} + \frac{1}{2}(1 + E_H) \\
\frac{1}{2}E_H &= 1 \\
E_H &= 2
\end{aligned}$$

Similarly since there is a 0.5 probability we get tails on the first flip, $E_T = E_H = 2$. Then,

$$\begin{aligned}
E_{HT} &= \frac{1}{2}(1 + E_H) + \frac{1}{2}(1 + E_T) \\
E_{HT} &= \frac{1}{2}(1 + 2) + \frac{1}{2}(1 + 2) \\
E_{HT} &= \frac{6}{2} = 3
\end{aligned}$$

So the expected value E_{HT} for getting a heads and a tails is 3 flips.

3. Let E_{HH} be the number of expected flips for two consecutive heads to appear, E_H be the expectation that another heads appears given the last flip was heads. Then,

$$E_{HH} = \frac{1}{2}(1 + E_H) + \frac{1}{2}(1 + E_{HH})$$

And E_H is,

$$E_H = \frac{1}{2} + \frac{1}{2}(1 + E_{HH})$$

Then,

$$\begin{aligned}
E_{HH} &= \frac{1}{2}(1 + E_H) + \frac{1}{2}(1 + E_{HH}) \\
E_{HH} &= \frac{1}{2}\left(1 + \left(\frac{1}{2} + \frac{1}{2}(1 + E_{HH})\right)\right) + \frac{1}{2}(1 + E_{HH}) \\
E_{HH} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}E_{HH} + \frac{1}{2} + \frac{1}{2}E_{HH} \\
E_{HH} &= \frac{3}{2} + \frac{3}{4}E_{HH} \\
\frac{1}{4}E_{HH} &= \frac{3}{2} \\
E_{HH} &= \frac{12}{2} = 6
\end{aligned}$$

So the expected value E_{HH} for getting two consecutive heads is 6 flips.